UNIT: II

Authentication of Electronic Records & E- Governance

❖ Authentication of Electronic Records

Authentication of electronic records ensures a record is a true and accurate representation of what it claims to be, often through digital signatures, which depend on asymmetric cryptography and hash functions. A subscriber's unique digital signature is created with a private key and verified with the corresponding public key, ensuring the record's origin and integrity. The process makes it computationally infeasible to alter the record without detection or to derive the original record from its hash result. Under Indian law, this is primarily governed by **Section 3 of the Information Technology Act, 2000**.

Key Concepts

- **Digital Signature:** A mathematical technique that provides the authenticity and integrity of an electronic record.
- **Public Key Infrastructure (PKI):** Involves public and private keys that are mathematically linked.
- **Hash Function:** An algorithm that takes input data and produces a fixed-size "hash result," or "message digest".
- **Private Key:** Used by the subscriber to create the digital signature.
- **Public Key:** Used by anyone to verify the digital signature and the authenticity of the record.

❖ Digital Signatures

A digital signature is a cryptographic method for validating digital documents, providing the highest level of assurance for authenticity, integrity, and non-

repudiation. It functions like a digital fingerprint, created using a private key, which confirms the signer's identity and ensures the document hasn't been tampered with. In many countries, digital signatures are legally required and equivalent to handwritten signatures.

Digital signatures use public-key cryptography (PKI) to provide security for digital information.

1. **Hash Generation:** A unique mathematical value called a hash is created from the digital document.
2. **Encryption:** The sender uses their private key to encrypt this hash, creating the digital signature.
3. **Verification:** The recipient uses the sender's public key to decrypt the signature, which is then compared to a new hash generated from the received document.

❖ RSA and Digital Signature

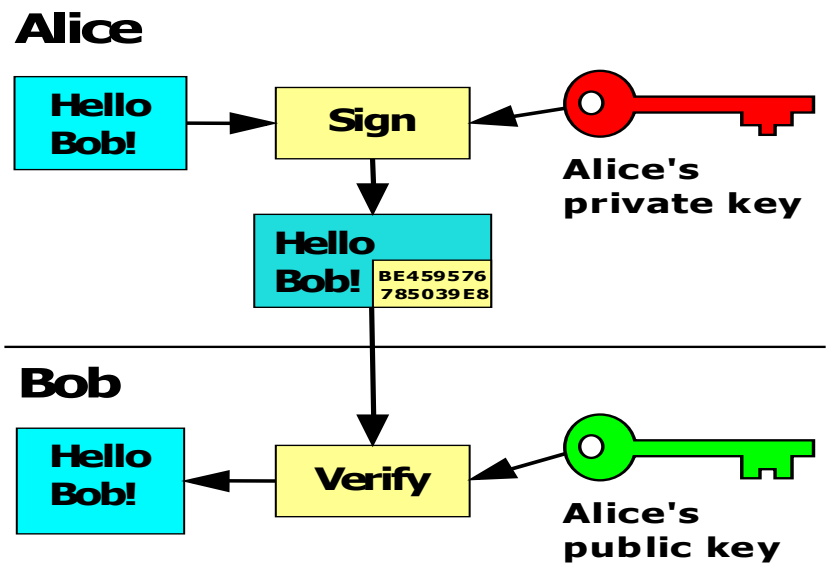
What is RSA: It is the most popular asymmetric cryptographic algorithm. It is primarily used for encrypting messages but can also be used for performing digital signatures over a message.

What is Digital Signature: As the name sounds are the new alternative to signing a document digitally. It ensures that the message is sent by the intended user without any tampering by any third party (attacker). In simple words, digital signatures are used to verify the authenticity of the message sent electronically.

RSA Signature

Let us understand how RSA can be used for performing **digital signatures step-by-step**. Assume that there is a sender (A) and a receiver (B). A wants to send a message (M) to B along with the digital signature (DS) calculated over the message.

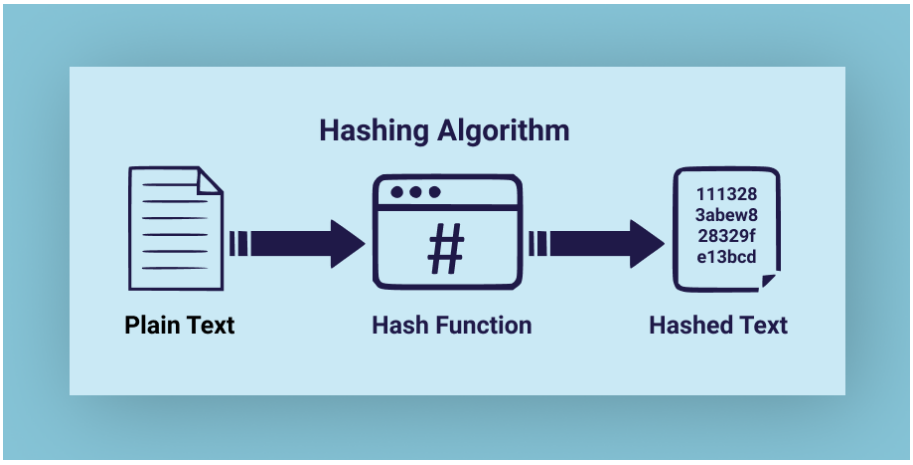
- **Step 1:** Sender A uses the SHA-1(Secure Hash Algorithm 1) Message Digest Algorithm to calculate the message digest (MD1) over the original message M.
- **Step 2 :** A now encrypts the message digest with its private key. The output of this process is called Digital Signature (DS) of A.
- **Step 3 :** Now sender A sends the digital signature (DS) along with the original message (M) to B.
- **Step 4 :** When B receives the Original Message(M) and the Digital Signature(DS) from A, it first uses the same message-digest algorithm as was used by A and calculates its own Message Digest (MD2) for M.
- **Step 5 :** Now B uses A's public key to decrypt the digital signature because it was encrypted by A's private key. The result of this process is the original Message Digest (MD1) which was calculated by A.



❖ Hash Function

A **hash function** is like a digital fingerprint generator—it takes any input (text, number, file, etc.) and produces a fixed-size string of characters, which is typically a sequence of numbers and letters. This output is called a **hash value** or **digest**.

A hash function is a mathematical algorithm that converts data of any size into a fixed-size string of characters called a hash value, or hash. This process is deterministic, meaning the same input always produces the same output. Hash functions are used for efficient data retrieval in hash tables and for security purposes in digital signatures and blockchains.



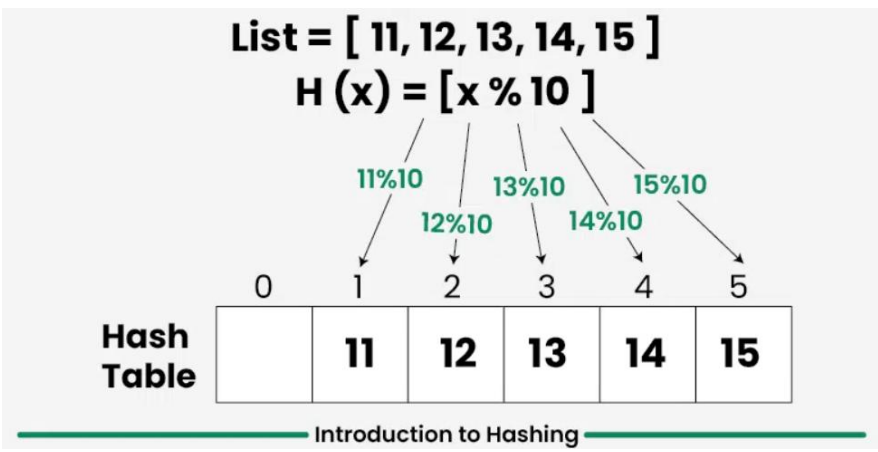
Key Properties of Hash Functions

- **Deterministic**: Same input always gives the same output.
- **Fixed Output Size**: No matter how big the input is, the hash is always the same length.
- **Fast Computation**: It should be quick to compute the hash.
- **Avalanche Effect**: A tiny change in input drastically changes the output.
- **One-Way**: You can't reverse-engineer the input from the hash.
- **Collision Resistant**: It’s hard to find two different inputs that produce the same hash.

Common Uses of Hash Functions

- **Hash Tables:** Hash functions map keys to indices in a hash table, enabling quick data storage and retrieval.
- **Digital Signatures:** By creating a hash of a message, hash functions ensure its integrity; if the message or hash is altered, the verification process fails.
- **Cryptocurrencies:** Hash functions are fundamental to blockchain technology, used to link blocks and verify transactions, such as in Bitcoin.
- **Data Integrity:** Hashes are used to verify that a file or message has not been tampered with by comparing its current hash to a previously stored one.

Hashing refers to the process of generating a small sized output (that can be used as index in a table) from an input of typically large and variable size. Hashing uses mathematical formulas known as hash functions to do the transformation. This technique determines an index or location for the storage of an item in a data structure called Hash Table.



❖ Working of Digital Signature

A **digital signature** is like a virtual seal of authenticity—it proves that a document or message was created by a specific person and hasn't been tampered with. It's built on **cryptography**, especially **asymmetric encryption**, which uses two keys: a **private key** (kept secret by the signer) and a **public key** (shared with everyone).

how it works step by step:

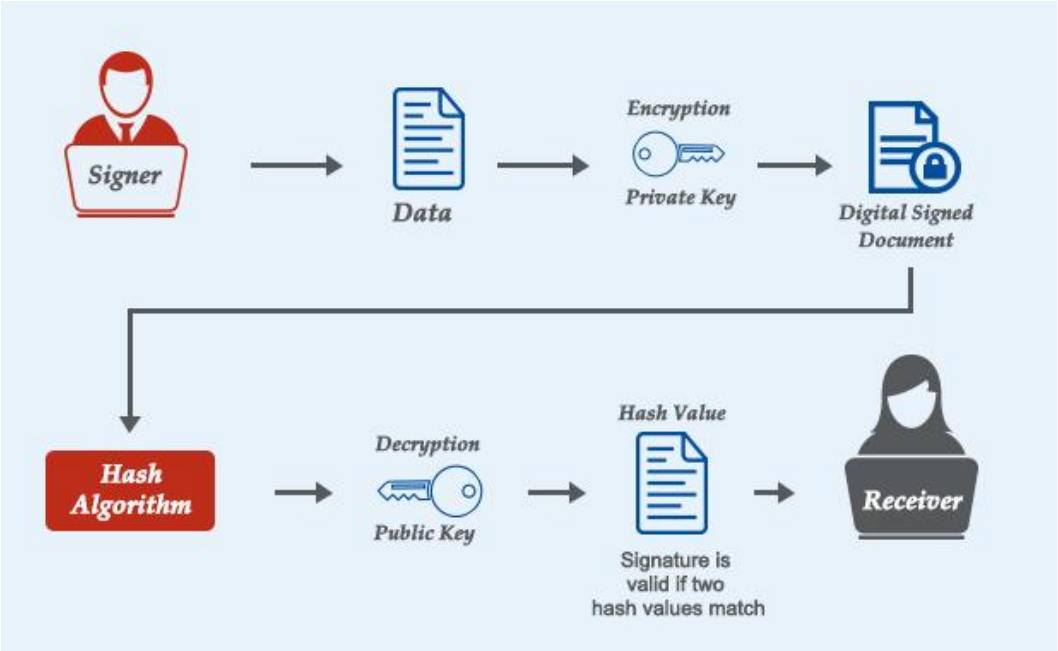
How a Digital Signature Is Created-

1. **Hashing the Document**
    - The document is passed through a **hash function** (like SHA-256), which generates a unique fixed-length string called a **message digest**.
    - This digest represents the content. Even a tiny change in the document will produce a completely different hash.
2. **Encrypting the Hash with Private Key**

- The sender uses their **private key** to encrypt the hash. This encrypted hash becomes the **digital signature**.
- The original document and the digital signature are then sent to the recipient.

How It's Verified-

1. **Decrypting the Signature**
    - The recipient uses the sender's **public key** to decrypt the digital signature. This reveals the original hash.
2. **Hashing the Received Document**
    - The recipient runs the received document through the same hash function to generate a new hash.

3. **Comparing Hashes**

- If the decrypted hash matches the newly generated hash, the document is **authentic** and **untampered**.
- If they don't match, it means the document was altered or the signature is invalid.